

LAB IV: SIMULATING INSTRUCTION CYCLES USING PYTHON

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Marks: ____/10

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Objectives:

1. To simulate the **Fetch Cycle and Decode Cycle** of a simple instruction execution process using Python.
2. To implement the **Fetch, Decode** and **Execute** cycles of a simple instruction processing system using Python.
3. To implement a **branch instruction** (BEQ – Branch if Equal to Zero) along with basic arithmetic and control flow using a Python-based processor simulation.

Post Lab:

1. *What are the three cycles of instruction execution?*
2. Difference between fetch and decode cycles.
3. What are the CPU registers?
4. *Why is the instruction cycle important in computer architecture?*
5. *Difference between JMP and JNZ instructions.*